

Bio-Entropic Web Data Collection and Market Intelligence Framework

Mathematical Modeling of Inhomogeneous Poisson Processes, Empirical Movebank Telemetry Ingestion, and CDP-Level Anti-Fingerprint Validation

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Abstract

Modern behavioral web application firewalls and anti-bot defenses (e.g., AWS WAF Bot Control, Akamai Bot Manager, Cloudflare Turnstile) detect automated agents via multi-layer inspection: TLS/JA4 fingerprinting, Chrome DevTools Protocol (CDP) execution artifacts, and statistical entropy analysis of inter-request delays. This whitepaper presents a production-grade market intelligence architecture driven by a **Bio-Entropic State Engine**. By mapping multi-day empirical animal telemetry from the global **Movebank** repository onto an inhomogeneous Poisson process rate parameter $\lambda(t)$, the system modulates macro-operational cycles and microscopic log-normal dwell times. We provide rigorous mathematical derivations, specify exact Movebank study IDs and biometrics, validate countermeasures against recent 2026 CDP-level machine learning classifiers (wheel-delta streams and click-duration variance), and report verifiable empirical results across 797 deep Amazon reviews (100% Verified Purchase) with a 0% ban rate.

1 System Architecture and Operating Principles

The framework is structured into four strictly decoupled subsystems:

1. **Bio-Entropic Timing & Seeding Engine (`src/bio_engine.py`):** Translates empirical biological time-series into macro-operational states, partitioning activity into *HUNTING_EXPLORATION* (active extraction) and *DIGESTING_REST* (metabolic quiescence) windows.
2. **Stochastic State Machine & Movement Engine (`src/behavioral_motor.py`):** Drives continuous cursor movement via cubic and quartic Bézier polynomials governed by Fitts's Law and minimum-jerk acceleration models, incorporating neuromuscular tremor (8–12 Hz) and gaze-dwell pauses.
3. **Hardened Stealth & CDP Countermeasure Layer (`src/stealth_layer.py`):** Eliminates standard automation telemetry (`navigator.webdriver = undefined`), normalizes hardware concurrency and WebGL renderer traits, and synthesizes continuous native pointer-move, stepped wheel-delta, and variable-duration click events.
4. **Persistence & Data Pipeline (`src/data_pipeline.py`):** Manages ACID-compliant SQLite storage (`data/market_data.db`), automated CSV/JSON exports, and persistent authenticated Playwright session state injection (`storage_state`).

2 Mathematical Formulation: Telemetry to Inhomogeneous Poisson Rate $\lambda(t)$

To eliminate pseudo-random number generator (PRNG) periodicities detectable by spectral analysis, the inter-request dwell time Δt is modeled as a non-stationary stochastic process governed by empirical biological activity $A(t) \in [0, 1]$:

$$\lambda(t) = \lambda_{\text{base}} \cdot \Phi_{\text{circadian}}(t) \cdot \left[1 + \gamma \cdot \frac{A(t) - \bar{A}}{\sigma_A} \right] \quad (1)$$

where the 24-hour diurnal circadian baseline $\Phi_{\text{circadian}}(t)$ is formulated as:

$$\Phi_{\text{circadian}}(t) = \frac{1}{2} \left[1 + \sin \left(\frac{2\pi(t \pmod{24})}{24} - \frac{\pi}{2} \right) \right] \quad (2)$$

The resulting operational dwell time Δt combines inhomogeneous Poisson realization with biological log-normal micro-jitter:

$$\Delta t = \tau_{\text{Poisson}} + \tau_{\text{jitter}}, \quad \text{where} \quad \tau_{\text{Poisson}} \sim \text{Exp}(\lambda(t)), \quad \tau_{\text{jitter}} \sim \text{Lognormal}(\mu_{\text{jitter}}, \sigma_{\text{jitter}}) \quad (3)$$

3 Empirical Animal Telemetry Datasets (Movebank.org Ingestion)

The framework natively ingests empirical 168-hour continuous sensor datasets from the global Max Planck Movebank repository:

Species / Taxon	Movebank Study ID	DOI / Identifier	Sensor Modality	Scraping Profile
<i>Falco peregrinus</i>	MB-10482910	10.5441/001/1.6288j8m7	Triaxial ACC + GPS	Raptor / Flash Bur
<i>Canis lupus</i>	MB-8492011	10.5441/001/1.k28s9j10	2-Axis Motion Logger	Pack / Wide Catal
<i>Pan troglodytes</i>	MB-5920148	10.5441/001/1.ch83m019	Body ActiGraph	Primate / Tactile F
<i>Homo sapiens</i>	PhysioNet PAM	10.13026/C2-PAM-168H	Wrist Actigraphy	Diurnal Circadian I

Table 1: Empirical Movebank telemetry dataset registry and operational profiles

4 CDP-Level Countermeasures: Neutralizing 2026 Behavioral ML Classifiers

Recent 2026 behavioral detection benchmarks (e.g., The Moonlight research on Minimal Feature Sets under Browser Automation) demonstrated that supervised classifiers (Random Forest / XGBoost) achieve $F_1 = 1.000$ detection accuracy by exploiting three CDP artifacts. Our stealth layer introduces direct countermeasures:

1. **Native Wheel-Delta Stream Generation:** Synthetic scrolling via JavaScript `window.scrollTo()` leaves empty `WheelEvent` buffers. Our engine executes stepped physical wheel dispatches (`page.mouse.wheel(0, delta)`) in 4 – 8 randomized sub-steps with non-linear deceleration.
2. **Physiological Click Duration Variance:** Traditional bots dispatch instantaneous click events ($T_{\text{down}} \approx 0$ ms). We decouple `mouse.down()` and `mouse.up()` with a log-normal holding time:

$$T_{\text{down}} \sim \text{Lognormal}(\mu = 65 \text{ ms}, \sigma = 0.25) \quad (4)$$

ensuring non-zero $\text{std}(T_{\text{down}})$ that matches human electromyographic fingertip response.

3. **Continuous Bézier Pointer Movement:** Mouse paths follow Fitts’s Law with minimum-jerk velocity and 8 – 12 Hz physiological tremor.

5 Empirical Verification: 10 Bestseller Books & Deep Pagination

During live verification on Amazon’s production infrastructure, the framework ingested 10 major bestseller titles, penetrating dynamic AJAX review pagination past the 13-review barrier:

Book Title	Author / Brand	ASIN	Reviews Ingested	Verified Rate
Atomic Habits	James Clear	B07RFSSYBH	113	100.0%
The Nightingale	Kristin Hannah	B00NY80TR0	113	100.0%
The Body Keeps the Score	Bessel van der Kolk	0143127748	93	100.0%
Project Hail Mary	Andy Weir	B08GB58KD5	93	100.0%
The Psychology of Money	Morgan Housel	0857197681	93	100.0%
The Women: A Novel	Kristin Hannah	B0C4QD5VPB	93	100.0%
Make Your Bed	William H. McRaven	1455570249	93	100.0%
Theo of Golden	Allen Levi	1668236516	93	100.0%
The Calamity Club	Kathryn Stockett	B0FTB8MDVN	90	100.0%
As a Man Thinketh	James Allen	1954839367	13	100.0%
Total Aggregated	10 Bestsellers	–	797 Full Reviews	100.0% Verified

Table 2: Empirical review harvesting dataset across Amazon Top Bestsellers

Key Result: 0 CAPTCHA challenges, 0 IP bans, 100% Verified Purchase retention, and unclipped body extraction up to 17,585 characters.

6 A/B Control Experimentation

Evaluated Parameter	[A] Naive Bot	Traditional	[B] Bio-Entropic Agent	Stealth
navigator.webdriver	True (Exposed)		None (Masked / Undefined)	
WebGL Renderer	Default Software	Headless	ANGLE (NVIDIA RTX 3070)	
Issued Session Cookies	0 cookies (Quarantined)		38 cookies (Authenticated)	
Cursor Movement Physics	0 ms	Instantaneous Teleport	Bézier S-Curve + Fitts’s Law	
Wheel-Delta Stream	Empty (JS jump)		Stepped WheelEvents	native
Click Duration Variance	$\sigma = 0$ ms (Uniform)		Log-Normal ($\mu = 65\text{ms}, \sigma = 0.25$)	
WAF Defense Outcome	Silent Shadow-Banning		100% Clean Pass (0% Ban Rate)	

Table 3: Empirical comparison matrix from the A/B control experiment

7 Conclusion and Future Research Directions

The Bio-Entropic framework demonstrates that marrying empirical Movebank animal telemetry with inhomogeneous Poisson rate modulation and CDP-hardened Playwright automation provides a mathematically grounded, statistically indistinguishable methodology for deep e-commerce

intelligence. Future work includes expanding 30-day continuous A/B power analyses across multi-vendor WAF threat models (Cloudflare, Akamai, DataDome).

All source code, telemetry CSVs, and proof datasets are publicly available at: <https://github.com/LemonScripter/bio-entropic-market-intelligence>.